GAME ART DIRECTION DOCUMENT

Sakartvelo Defenders

Art Style Guide v2.1
Blender Pipeline Edition

2D Isometric Cel-Shaded Art Direction with 10-Era Color Palette System, Blender 3D-to-2D Pipeline, Tower Silhouette System, and Character Proportions. Aligned with Game Design Bible v3.0.

VERSION	DATE	ENGINE
v2.1	April 2026	Godot 4.x + Blender

CONFIDENTIAL ART PRODUCTION REFERENCE

Table of Contents

1. Art Direction 5

1.1 Overall Style: 2D Isometric Cel-Shaded 5

1.2 Production Pipeline: Blender to 2D Sprites 5

1.3 Rendering Rules (Visual Output) 5

2. Global UI Palette 6

3. Era Color Palettes 7

3.1 Era 0: Ancient Colchis 8

3.2 Era 1: Kingdom of Iberia 8

3.3 Era 2: Age of Invasions 8

3.4 Era 3: The Georgian Golden Age 8

3.5 Era 4: The Mongol Catastrophe 9

3.6 Era 5: Between Empires 9

3.7 Era 6: Russian Empire 9

3.8 Era 7: First Democratic Republic 9

3.9 Era 8: The Soviet Century 9

3.10 Era 9: Modern Georgia 10

4. Tower Silhouette Guidelines 10

4.1 Silhouette Reference 10

4.2 Silhouette Rules 11

5. Character Proportion Reference 11

5.1 Standard Proportions (10-Unit System) 11

5.2 Proportion Variations by Character Type 12

6. Hero Portrait Guidelines 12

6.1 Composition Rules 12

6.2 Portrait Render Setup 13

7. Enemy Design Guidelines 13

7.1 Enemy Category Visual Language 13

7.2 Enemy Color Rules 13

7.3 Cultural Sensitivity Standards 14

8. Map Design Guidelines 14

8.1 Map Composition 14

8.2 Elevation Rendering 14

8.3 Era Map Themes 15

9. UI Design Language 15

9.1 UI Components 15

9.2 HUD Layout 16

10. AI Concept Reference System 16

10.1 Prompt Architecture 16

10.2 Prompt Keywords by Era 17

10.3 Concept-to-Model Workflow 17

11. Blender Pipeline 18

11.1 Scene Setup 18

11.2 Material System 18

11.3 Outline Rendering (Freestyle) 19

11.4 Polycount Budgets 19

11.5 Sprite Sheet Export 20

11.5.1 Animation Frames 20

11.5.2 Render Resolution 20

11.5.3 Sprite Sheet Layout 20

11.6 Blender Project Organization 21

12. Scroll and Mastery Visual Design 21

12.1 Scroll Appearance 21

12.2 Mastery Badge Display 22

13. File Naming Convention 22

13.1 Blender Source Files 22

13.2 Rendered Sprite Files 22

13.3 Material and Palette Files 22

13.4 Naming Rules 23

14. Consistency Checklist 23

14.1 Visual Consistency 23

14.2 Pipeline Consistency 23

15. Hermes Collaboration Scope 24

15.1 Hermes Responsibilities 24

15.2 Hermes Limitations 24

15.3 Priority Work Order 25

16. Version History 25

v2.1 - Blender Pipeline Update 25

v2.0 - Initial Release 26

1. Art Direction

1.1 Overall Style: 2D Isometric Cel-Shaded

Sakartvelo Defenders uses a 2D isometric art style with cel-shaded lighting. All game assets, including towers, enemies, heroes, maps, and UI elements, share this foundational visual language. The isometric perspective provides clear spatial awareness for tower placement decisions while maintaining a warm, illustrated quality that suits the historical subject matter. Cel-shaded lighting creates sharp shadow boundaries and flat color fills that give the art a distinctive, hand-illustrated appearance that complements the historical theme.

The art style draws inspiration from illustrated history books, museum dioramas, and traditional Georgian miniature painting. Every asset should feel like it belongs in a high-quality history textbook rather than a generic mobile game. The color palette for each era is carefully curated to reflect the historical period's visual character, while maintaining overall cohesion through shared rendering techniques including consistent shadow direction, outline weight, and texture density across all assets.

1.2 Production Pipeline: Blender to 2D Sprites

All game art assets are produced using Blender as the primary 3D modeling and rendering tool, then exported as 2D sprite sheets for use in the Godot 4.x game engine. The game displays these assets as flat 2D images in an isometric perspective; there is no real-time 3D rendering during gameplay. The 3D production pipeline provides several critical advantages over direct 2D illustration or AI generation: perfectly consistent lighting and shadows across all assets, precise control over proportions through shared armatures and reference rigs, efficient iteration through material and texture swaps without re-modeling, and automatic generation of rotation angles and animation frames from a single model.

The workflow follows a strict linear pipeline:

- Concept Reference (AI-generated mood boards, historical photographs, archaeological references)
- 3D Modeling (Blender, low-poly models optimized for isometric rendering)
- Texturing (flat color materials matching era palettes, cel-shader node setup)
- Rendering (orthographic camera, Freestyle outlines, sprite sheet export)
- Engine Integration (Godot 4.x sprite import, animation setup)

Each stage has defined quality gates documented in the Blender Pipeline section (Section 11). AI image generation is used exclusively during the Concept Reference stage for mood boards and visual exploration, never as final production output.

1.3 Rendering Rules (Visual Output)

These rules define the final rendered appearance of all 2D sprites in the game. Whether an asset is a tower, enemy, hero, or environmental prop, the rendered output must conform to these specifications.

These are the rules that the Blender rendering pipeline must reproduce consistently across every asset and every era.

Shadow Direction: All shadows cast to the lower-right at a 45-degree angle, consistent with the isometric camera position. Shadow color is a desaturated version of the object's base color at 40% opacity. Shadow offset is proportional to object height: taller objects cast longer shadows. In Blender, this is achieved with a single directional light (Sun type) rotated to -45 degrees on the X axis and 45 degrees on the Y axis from the orthographic camera.

Outline Weight: All objects have a 2px dark outline (#1A1A1A) on all visible edges except where objects contact the ground plane. Ground shadows do not have outlines. Character outlines are slightly thicker (3px) for readability at small sizes. In Blender, outlines are produced using Freestyle edge rendering with line thickness calibrated per object category.

Color Fills: Flat color fills with no gradients except for metallic surfaces (swords, armor) which use a simple two-tone highlight/shadow gradient. Color values must match the era-specific palette exactly; no deviations are permitted without explicit approval from the style guide. In Blender, materials use Principled BSDF with roughness set to 1.0 for non-metallic surfaces.

Texture: Minimal texture usage. Stone surfaces may include a subtle noise overlay at 5-10% opacity. Wood grain is represented by 2-3 parallel lines rather than photorealistic texture. Fabric is flat colored with fold lines only on capes and robes. Normal maps are not used.

Lighting Model: Single directional light source from the upper-left, creating consistent highlights on all surfaces. The highlight color is the era's highlight palette color at 30% opacity, blended additively. No ambient occlusion or global illumination effects.

Anti-Aliasing: All renders use 8x multi-sample anti-aliasing to ensure smooth edges at all output resolutions. No post-processing effects (bloom, depth of field, motion blur) are applied.

2. Global UI Palette

The global UI palette is used for all user interface elements that are not era-specific: menus, buttons, HUD elements, settings panels, the marketplace, and loading screens. Era-specific content uses the era palettes defined in Section 3. The global UI palette is designed to be neutral and readable, complementing any era's color scheme without clashing.

Element	Color(s)	Usage
Background	#1A1A1A / #F5F0EB	Dark mode default; light mode in settings
Primary Text	#FFFFFF / #1A1A1A	White on dark / Black on light
Secondary Text	#B0B8C0 / #6B7280	Muted labels and descriptions
Accent	#E67E22	Interactive elements, selected states, CTAs

Accent Hover	#F39C12	Button hover state
Danger	#E74C3C	Damage indicators, low-health warnings
Success	#27AE60	Level completion, achievement unlocks
Info	#3498DB	Tooltips and educational pop-ups
Panel Background	#2C3E50 / #FFFFFF	Cards, modals, overlay panels
Border	#3D566E / #E5E7EB	Subtle borders between UI sections
Disabled	#4A5568 / #D1D5DB	Inactive buttons, locked content

Table 2.1: Global UI Palette

3. Era Color Palettes

Each of the 10 eras has a dedicated 8-color palette that defines the visual character of all era-specific assets. These palettes are derived from historical research into the architecture, clothing, military equipment, and natural environment of each period. The 8 color slots serve specific roles: Base (dominant color for large surfaces), Highlight (brightest color, light-facing planes), Shadow (darkest color, depth), Accent (era signature color), Sky (background gradient), Vegetation (trees, grass, crops), Stone/Earth (building materials), Water (rivers, fountains, coast).

Era	Name	Base	Highlight	Shadow	Accent	Sky	Veg	Stone	Water
0	Ancient Colchis	#2D5A3D	#4CAF50	#1B3A26	#D4A017	#87CEEB	#3A7D44	#8B7355	#2E86AB
1	Kingdom of Iberia	#4A3728	#8D6E63	#2C1F17	#C8A96E	#7FB3D8	#5D8A3C	#9E8B6E	#4682B4
2	Age of Invasions	#5C3A21	#A0522D	#3B2510	#DAA520	#B8860B	#6B8E23	#8B7D6B	#4A708B
3	Georgian Golden Age	#1A4D2E	#D4AF37	#0D2818	#C41E3A	#4A90D9	#4CAF50	#B8860B	#2980B9
4	Mongol Catastrophe	#3D2B1F	#8B6914	#261A11	#CC5500	#8B4513	#556B2F	#696969	#4682B4
5	Between Empires	#1A3C5E	#3498DB	#0D2137	#E74C3C	#F5DEB3	#2E8B57	#CD853F	#5F9EA0
6	Russian Empire	#2C2C3E	#7B8DB1	#1A1A2E	#9B59B6	#BDC3C7	#4A7C59	#8B8682	#5B8FA8
7	First Dem. Republic	#1B4332	#52B788	#0B2618	#E63946	#A8DADC	#40916C	#6B705C	#457B9D

8	The Soviet Century	#2D2D2D	#757575	#1A1A1A	#D32F2F	#90A4AE	#558B2F	#795548	#546E7A
9	Modern Georgia	#0D1B2A	#415A77	#070E18	#E63946	#E0E1DD	#2D6A4F	#777777	#1D3557

Table 3.0: Complete Era Palette Reference

3.1 Era 0: Ancient Colchis

The Colchis palette is dominated by deep forest greens and rich earth tones, reflecting the lush Black Sea coastal environment described in Greek sources. The golden accent (#D4A017) references the Golden Fleece legend and the historical gold-mining activities of the Colchian civilization. The base green (#2D5A3D) evokes the dense Colchian forests that were a major strategic asset in ancient warfare, providing timber for fortifications and concealment for defenders. Tower skins in this era use the Stone/Earth palette for foundations and the Base green for wooden superstructures. The Accent gold is used for decorative elements on important structures and for Medea's hero design. Enemy designs for tribal raiders use the Shadow palette for leather armor and the Accent gold for jewelry and ceremonial items.

3.2 Era 1: Kingdom of Iberia

The Iberian palette shifts to warm browns and muted golds, reflecting the stone architecture and arid highland landscapes of early Georgian statehood. The period covers the transition from pagan to Christian Georgia, and the palette balances the warm earth tones of pre-Christian architecture with the nascent use of cross symbols in decorative elements. The highlight brown (#8D6E63) represents the sandstone used in Iberian fortresses, while the accent gold (#C8A96E) evokes the royal Bagrationi regalia that would become iconic in later eras. Tower skins in this era feature more substantial stone construction compared to Era 0. The introduction of Christianity is reflected in the subtle use of cross motifs on tower decorations.

3.3 Era 2: Age of Invasions

The Age of Invasions palette is deliberately harsh and desaturated, reflecting the devastation of the Arab conquest period and the ongoing military pressure on Georgian lands. The base brown (#5C3A21) is darker and more austere than Era 1, representing fortresses under siege and a civilization under threat. The accent gold (#DAA520) is muted and battle-worn, symbolizing the resilience of Georgian culture under occupation rather than its prosperity. This era's visual language is defined by contrast between the worn, battle-scarred Georgian architecture and the imposing, geometric forms of Arab military architecture.

3.4 Era 3: The Georgian Golden Age

The Golden Age palette is the richest and most vibrant in the game, reflecting the historical period of Georgia's greatest power and cultural achievement. The deep green base (#1A4D2E) represents the lush Georgian countryside during a period of agricultural prosperity. The accent gold (#D4AF37) is pure and brilliant, symbolizing the legendary wealth of Queen Tamar's court. The accent red (#C41E3A) represents both the Georgian cross and the military banners of the unified Georgian army. Tower skins in this era are the most ornate in the game, featuring intricate stone carvings, gold decorative elements, and the distinctive Georgian cross motif.

3.5 Era 4: The Mongol Catastrophe

The Mongol Catastrophe palette is the darkest and most somber in the game, reflecting the destruction and despair of the Mongol conquest period. The base brown (#3D2B1F) is nearly black, representing burned and abandoned cities. The accent orange (#CC5500) is harsh and industrial, evoking fire and destruction rather than the warm gold of earlier eras. The vegetation color (#556B2F) is desaturated, suggesting scorched earth and dying crops. Visual storytelling in this era emphasizes contrast between the ruined Georgian architecture and the imposing, utilitarian forms of Mongol military camps.

3.6 Era 5: Between Empires

The Between Empires palette reflects the cosmopolitan character of Georgian culture during the Ottoman-Persian rivalry period. The blue base (#1A3C5E) represents the expanding influence of both empires on Georgian territory. The accent red (#E74C3C) is vivid and dramatic, symbolizing both the blood of constant warfare and the rich Persian-influenced textile traditions. This era's visual language is defined by cultural fusion. Tower skins feature a blend of Georgian, Persian, and Ottoman architectural elements.

3.7 Era 6: Russian Empire

The Russian Empire palette is cool and institutional, reflecting the period of annexation and the imposed European architectural influence on Georgian cities. The purple-gray base (#2C2C3E) represents the bureaucratic and melancholic character of the era, while the accent purple (#9B59B6) subtly references the Russian imperial aesthetic. Tower skins in this era mark the transition from traditional Georgian military architecture to European-influenced fortifications.

3.8 Era 7: First Democratic Republic

The First Republic palette is surprisingly optimistic, using fresh greens and blues to reflect the hope and democratic aspirations of Georgia's brief independence period (1918-1921). The base green (#1B4332) is lighter and more vibrant than earlier military eras, suggesting renewal and national rebirth. The accent red (#E63946) represents both the revolutionary spirit and the sacrifice of the independence wars.

3.9 Era 8: The Soviet Century

The Soviet Century palette is stark and utilitarian, dominated by grays and the iconic Soviet red. The base gray (#2D2D2D) represents the concrete brutalism of Soviet architecture, while the accent red (#D32F2F) is the unmistakable symbol of Soviet power. The desaturated blue-gray sky (#90A4AE) conveys the oppressive atmosphere of the period. Educational content for this era is presented with factual neutrality, and the visual design avoids both glorification and demonization.

3.10 Era 9: Modern Georgia

The Modern Georgia palette is dark and sophisticated, using deep navy blues and strategic pops of red to convey a contemporary, forward-looking aesthetic. The base navy (#0D1B2A) represents the night-time urban environment of modern Tbilisi, while the accent red (#E63946) connects to the Georgian flag and the ongoing national identity. Tower skins in this era feature cutting-edge military technology: precision-guided weapon systems, drone surveillance platforms, and fortified military outposts with modern communications equipment.

4. Tower Silhouette Guidelines

Each of the 10 tower types has a unique silhouette that must be recognizable at small sizes (approximately 64x64 pixels on mobile, 128x128 on desktop) without relying on color or detail. Silhouette distinctiveness is critical for gameplay readability: players must be able to identify tower types instantly during combat.

4.1 Silhouette Reference

Tower Type	Silhouette Shape	Height	Width	Key Identifier
Archer Tower	Tall narrow spire	8-10	3-4	Pointed roof, flag pole
Catapult/Trebuche t	Wide base, elevated arm	5-6	6-7	Pivoting arm, bucket
Wall/Barrier	Low wide rectangle	2-3	6-8	Crenellations, flat top
Cavalry Outpost	Medium organic horse	5-6	5-6	Horse head, rider
Siege Tower	Tall boxy w/ battlements	9-10	5-6	Square top, arrow slits
Magic/Shrine	Domed, ornate w/ spire	7-8	4-5	Dome, cross/crescent top
Gunpowder Tower	Medium cylindrical	5-6	4-5	Cannon barrel protrusion
Industrial Tower	Tall mechanical	8-9	4-5	Chimney, gear mechanisms
Bunker Tower	Low flat angular	3-4	5-6	Slotted window, angled walls
Tech Tower	Sleek angular	6-7	3-4	Dish/antenna, LED elements

Table 4.1: Tower Silhouette Reference

4.2 Silhouette Rules

No Overlap: No two tower types should share the same basic silhouette shape. If two towers appear identical in silhouette, the design must be revised. In Blender, silhouette comparison is performed by rendering all towers as flat black shapes against a white background and checking for visual confusion.

Scale Consistency: All towers are designed on the same grid scale. A 10-unit tall Archer Tower and a 3-unit tall Wall/Barrier must appear correctly proportioned when placed next to each other. The Blender unit system uses 1 Blender unit = 1 game unit for direct proportion mapping.

Era Skin Layering: Era-specific skins are applied as a material and geometry overlay on the base model. The silhouette shape never changes across eras; only the surface materials, decorative geometry, and colors vary.

Readability at Small Size: All tower designs must be tested at 64x64 pixel resolution in the Godot engine. If a tower cannot be identified at this size, the design is too complex and must be simplified.

Upgrade Visual Feedback: Tower upgrades (Level 1 to Level 3) must be visually distinguishable. Level 2 adds minor decorative elements. Level 3 adds significant visual changes: additional height, new structural elements, or glowing emission materials.

5. Character Proportion Reference

All humanoid characters (heroes, enemies, NPCs) follow a standardized 10-unit proportion system. This system ensures visual consistency across all character models and provides clear guidelines for Blender modeling. The 10-unit system divides the character's total height into 10 equal Blender units, with specific body parts occupying defined proportions.

5.1 Standard Proportions (10-Unit System)

Unit	Body Part	Proportion	Notes
1-2	Head	2 units (20%)	Larger than realistic for readability at small sizes
3	Neck + Shoulders	1 unit (10%)	Broad shoulders for heroes, narrower for civilians
4-5	Torso (upper)	2 units (20%)	Armor/clothing detail zone
6	Torso (lower) + Belt	1 unit (10%)	Belt/buckle as era identifier
7-8	Legs (upper)	2 units (20%)	Armor/clothing varies by era
9-10	Legs (lower) + Feet	2 units (20%)	Footwear reflects era technology

Table 5.1: Standard 10-Unit Proportions

5.2 Proportion Variations by Character Type

Heroes: 11 units total. Head is 2.5 units for greater facial detail visibility. Shoulders are 1.2 units wider than standard. Torso is elongated by 0.5 units to convey authority. These subtle modifications make heroes visually distinct from standard units without breaking the overall proportion system.

Boss Enemies: 12-14 units total. Entire body scaled up uniformly by 20-40%. This uniform scaling ensures that boss silhouettes are immediately recognizable as larger versions of their unit type. Boss proportions follow the same 10-unit ratios applied to the larger frame.

Cavalry Units: 12 units total (10 for rider + 2 for horse head). Horse body extends 3 units behind the rider's center point. Cavalry proportions only apply to the rider; the horse follows separate animal proportion guidelines.

Flying Units: 8 units total (more compact to suggest agility). Wings extend 3-4 units from the shoulder line on each side. Wing membranes use the era's Shadow color at 60% opacity for a translucent effect, achieved through Blender alpha blend mode on the wing material.

6. Hero Portrait Guidelines

Hero portraits are the most detailed character art in the game, appearing in the hero selection screen, level loading screens, and the collection gallery. Each portrait is a half-body or three-quarter view composition rendered from the hero's 3D model with a dedicated portrait camera setup. Portraits must work at both large sizes (gallery view) and small sizes (selection thumbnails, 80x80 pixels).

6.1 Composition Rules

Framing: Half-body composition, cropped at the waist. The hero's face occupies the upper third of the portrait, centered horizontally. The hero's body is angled 15-30 degrees from the camera (three-quarter view), never a flat profile or direct front-facing pose.

Background: A blurred, desaturated version of the hero's era palette as a gradient. The hero's Accent color appears as a subtle radial glow behind the head, creating visual separation from the background without adding a visible halo effect.

Expression: Each hero has a defined expression that reflects their personality. Military heroes have determined, focused expressions. Cultural heroes have thoughtful, contemplative expressions. Support heroes have calm, serene expressions. No hero has an aggressive or angry expression; even in a war game, the tone is dignified rather than hostile.

Era Markers: Each portrait includes 2-3 small era-identifying elements in the background or clothing details: a cross for Christian-era heroes, a crown for royalty, a manuscript for literary figures, military insignia for commanders. These markers must be subtle enough to not distract from the portrait but clear

enough to communicate the hero's era at a glance.

6.2 Portrait Render Setup

Each hero portrait is rendered from a dedicated Blender scene using the following standardized setup. The portrait camera uses a 50mm focal length lens for natural perspective, positioned at eye level with the hero angled 20 degrees from center. Lighting uses the same single-directional Sun light as gameplay assets, supplemented by a very subtle fill light at 10% intensity to soften facial shadows. The render resolution is 512x512 pixels for the source asset, which Godot downscales for thumbnails. The background plane uses the era palette gradient with a Gaussian blur node in the compositor.

7. Enemy Design Guidelines

Enemy design follows the same 10-unit proportion system as heroes but with distinct visual language that makes enemies immediately distinguishable from player units at a glance. The enemy design philosophy is based on three principles: historical authenticity, mechanical clarity, and cultural sensitivity.

7.1 Enemy Category Visual Language

Category	Body Type	Movement	Color Treatment	Silhouette Key
Infantry	Standard 10-unit	Ground path	Era shadow + base	Shield/spear shape
Cavalry	10-unit + mount	Fast ground	Era base + highlight	Horse silhouette
Siege	Boxy, 6-8 units wide	Slow ground	Era stone + shadow	Wheeled/wide frame
Flying	Compact 8-unit	Air path	Era shadow, translucent wings	Wing spread shape
Boss	12-14 unit scale	Ground (varies)	Full era palette + glow	Unique per boss

Table 7.1: Enemy Category Visual Language

7.2 Enemy Color Rules

Enemies use a deliberately restricted palette subset to create visual contrast with player towers. While towers use the full 8-color era palette, enemies are rendered primarily in the Shadow and Stone/Earth colors of their era, with the Accent color used sparingly for shields, banners, and weapons. This color restriction ensures that player towers are visually dominant on the battlefield, making it easy to distinguish defensive structures from enemy units at a glance.

Enemy faction colors provide additional differentiation within each era. When an era features multiple enemy factions (such as Ottoman and Persian forces in Era 5), each faction receives a subtle color tint: Ottoman forces lean toward the blue end of the palette range, while Persian forces lean toward the red end. This faction tinting is achieved by shifting the Base and Stone/Earth materials by 15-20 hue degrees in the Blender color ramp.

7.3 Cultural Sensitivity Standards

Enemy designs must never use exaggerated ethnic features, caricatured facial structures, or stereotypical clothing that could be interpreted as mocking a real cultural group. Enemies are military units, depicted with the same dignity and accuracy as player units. Historical accuracy means showing real military equipment and tactics of the period, not reinforcing ethnic stereotypes. This principle is especially important for eras involving ongoing political sensitivities (Era 8 Soviet period, Era 9 modern conflicts) where enemy designs must be factually accurate and neutrally presented. All enemy models must pass a cultural sensitivity review before being approved for production rendering.

8. Map Design Guidelines

Maps are the visual backdrop for gameplay, providing the context for tower placement, enemy pathing, and strategic decision-making. Each of the 200 levels has a unique map, but maps within each era share visual characteristics defined by the era palette and the map design rules below. Maps must be visually interesting without distracting from the gameplay elements.

8.1 Map Composition

Layered Depth: Maps use three depth layers: foreground (interactive zone where towers are placed), midground (enemy path with terrain features), and background (sky, distant mountains, horizon). The foreground occupies the bottom 40% of the screen, the midground the middle 35%, and the background the top 25%.

Path Visibility: Enemy paths must be clearly visible against the terrain. Paths are rendered as lighter or darker strips that contrast with the surrounding vegetation. Path width is consistent within a map but may vary between maps. Paths never overlap or cross at the same elevation.

Placement Zones: Tower placement zones are indicated by subtle grid markers on the foreground terrain. These markers are visible enough to guide placement but not so prominent that they dominate the visual composition. Grid markers use the era's Shadow color at 30% opacity.

8.2 Elevation Rendering

Maps use visual elevation cues to communicate the height mechanic. Elevated positions are rendered with a visible base/foundation that shows the height difference from the ground level. The foundation uses the era's Stone/Earth color, and the elevated platform uses the era's Base color. Towers placed on

elevated positions appear visually higher on the screen, with a visible shadow beneath them indicating their height advantage. Elevation levels follow a simple two-tier system: ground level and elevated. Maps typically have 2-4 elevated positions, providing meaningful but not overwhelming strategic options.

8.3 Era Map Themes

Era	Primary Environment	Key Terrain Features	Weather/Atmosphere
0	Colchian forests, Black Sea coast	Dense forests, river crossings, gold mines	Misty, humid, golden light
1	Caucasian highlands, early cities	Mountain passes, cave cities (Uplistsikhe)	Crisp mountain air, bright sunlight
2	Fortified borderlands	Siege-damaged fortresses, mountain passes	Dusty, stormy, smoke from burning cities
3	Prosperous towns, grand castles	Cathedrals, monasteries (Gelati, Vardzia)	Warm golden light, clear skies
4	Devastated countryside	Burned cities, abandoned villages	Overcast, ash-choked, red sky
5	Three-kingdom diverse terrain	Wine country, fortified cities	Warm Mediterranean, dusty campaigns
6	Modernizing Tbilisi	Railway lines, European-style buildings	Industrial haze, mixed old/new
7	War-torn countryside	Trenches, barricaded streets	Dramatic clouds, tension lighting
8	Soviet industrial zones	Concrete blocks, factory complexes	Gray overcast, utilitarian
9	Modern Tbilisi, conflict zones	Contemporary military installations	Night-time urban, digital displays

Table 8.1: Era Map Themes

9. UI Design Language

The UI design language defines the visual treatment of all in-game interface elements: buttons, panels, health bars, resource counters, hero ability cooldowns, and notification systems. The UI language is era-independent (always using the Global UI Palette from Section 2) but adapts its border style and background opacity to complement the current era's visual character.

9.1 UI Components

Buttons: Rounded rectangles with 8px border radius. Primary buttons use Accent fill (#E67E22) with white text. Secondary buttons use Panel Background fill with Primary text. All buttons have a 2px

border in Border color. Hover state brightens fill by 15%. Press state darkens fill by 10% and shifts content 1px down. Disabled state uses Disabled fill with Disabled text.

Panels: Rounded rectangles with 12px border radius. Panel Background fill with Border color stroke. Panels have a subtle inner shadow (2px, 10% opacity) to create depth. Panels containing era-specific content include a thin accent strip (3px) at the top edge using the current era's Accent color.

Health Bars: Horizontal bars with 4px height, rounded ends. Full health uses Success green (#27AE60). Mid health (50-80%) uses Info blue (#3498DB). Low health (under 50%) uses Danger red (#E74C3C). Background bar uses Disabled color. Health bars include a 1px dark outline for readability.

Resource Counters: Icon + number combination, right-aligned in the HUD. Resource icons use 24x24px sprites. Numbers use Primary Text color with a 1px dark shadow for readability. Values update with a smooth count-up animation (0.3s duration).

Cooldown Indicators: Circular progress overlay on hero ability icons. The unfilled portion uses Disabled color at 60% opacity, while the filled portion uses the Info blue. A numerical countdown appears in the center when cooldown exceeds 3 seconds.

9.2 HUD Layout

The in-game HUD occupies fixed positions on the screen and must never overlap with tower placement zones or enemy paths. The top bar contains: resource counters (left), era/level indicator (center), and pause/settings buttons (right). The bottom bar contains: hero portrait and ability buttons (left), tower build menu (center), and wave/score information (right). All HUD elements use the Global UI Palette and have a semi-transparent Panel Background with Border stroke to ensure readability against any map background.

The HUD adapts minimally to screen size: on mobile (portrait), the tower build menu moves to the right edge as a vertical scrollable list. On desktop (landscape), it remains as a horizontal bar at the bottom.

10. AI Concept Reference System

AI image generation tools are used exclusively during the Concept Reference stage of the production pipeline, never as final production output. Their role is to generate mood boards, explore visual directions, and provide reference material for Blender 3D modeling.

10.1 Prompt Architecture

All AI concept prompts follow a standardized structure to ensure the generated reference images align with the era palettes and visual direction established in this guide:

```
[era name] Georgian [subject], [environment], [mood/atmosphere], historical
reference, concept art, isometric perspective, cel-shaded style, [era base
color] dominant palette, [era accent color] highlights, illustrated history
book quality, clean composition, no text, no watermark
```

Example (Era 3 Concept): "Georgian Golden Age castle fortress, Caucasus mountains background, warm golden light, prosperous medieval kingdom, concept art, isometric perspective, cel-shaded style, deep green dominant palette, gold and red highlights, illustrated history book quality, clean composition"

10.2 Prompt Keywords by Era

Era	Subject Keywords	Atmosphere Keywords	Material Keywords
0	Colchian, ancient, mythological, Bronze Age	Misty, lush, golden, river delta	Wood, bronze, gold, stone, animal hide
1	Iberian, early Christian, Roman-Parthian era	Mountainous, sacred, transitional	Stone, wood, iron, early brick, mosaic
2	Arab conquest, fortified, besieged	Harsh, dusty, war-torn, defiant	Dark stone, weathered wood, iron, leather
3	Golden Age, medieval Georgian, cathedral	Prosperous, divine, triumphant, golden	Ornate stone, gold leaf, silk, fresco
4	Mongol invasion, devastation, burning	Ash-filled, dark, desperate, scorched	Burned wood, cracked stone, ash, mud
5	Ottoman-Persian, early modern, wine country	Cosmopolitan, tense, cultural fusion	Brick, tile, bronze cannon, silk, wool
6	Russian imperial, industrializing, European	Institutional, melancholic, transitional	Brick, iron, glass, railway steel
7	Democratic republic, independence, 1918-1921	Hopeful, fragile, revolutionary, brief	Concrete, sandbag, early steel, canvas
8	Soviet, brutalist, industrial, WWII	Gray, oppressive, utilitarian, controlled	Reinforced concrete, steel, red paint
9	Modern Georgia, contemporary, high-tech	Sleek, nighttime, digital, resilient	Glass, steel, composite, LED, concrete

Table 10.1: Prompt Keywords by Era

10.3 Concept-to-Model Workflow

The workflow from AI concept reference to Blender model follows a structured process:

- Generate 4-8 concept images per asset using the prompt architecture above.
- Select the best composition and design elements from the generated images, noting specific features to incorporate into the 3D model.

- Create a concept brief document that lists the approved design elements, palette assignments, and key reference images.
- Model in Blender using the concept brief as the guide, not the raw AI images. This ensures the final 3D model serves the game's visual language rather than being constrained by AI generation limitations.

AI concept images are stored in the project's reference folder organized by era and asset type. They are never used directly in game builds or marketing materials. Their sole purpose is to inform 3D modeling decisions and provide visual direction for the Blender artist.

11. Blender Pipeline

This section defines the complete Blender production pipeline for Sakartvelo Defenders. All game assets are produced through this pipeline, which is designed for a solo developer workflow, prioritizing consistency, reusability, and efficiency over cinematic quality.

11.1 Scene Setup

Camera: Orthographic camera positioned for isometric view. Camera rotation: X = 60.0 degrees, Y = 0.0 degrees, Z = 45.0 degrees. This produces the standard 2:1 isometric pixel ratio. Camera scale adjusts per asset type to fit the asset within the render frame while maintaining consistent unit-to-pixel ratio across all assets.

Lighting: Single Sun light with Direction = (-1, -1, -1) normalized to produce upper-left illumination with lower-right shadows. Strength = 1.0. Color = White (#FFFFFF). No fill lights, no rim lights, no HDRI environment lighting. A single ambient light at 0.15 intensity prevents completely black shadow areas.

Background: Plain color (#808080) for transparency compositing, or a gradient backdrop plane for portrait renders. The render output uses PNG with alpha channel for all gameplay assets. Map backgrounds use a separate opaque render with the era's Sky gradient.

11.2 Material System

All materials in the project use a standardized cel-shading material setup, designed for simplicity and consistency, using flat colors with sharp shadow boundaries.

Base Material (Non-Metallic): Principled BSDF shader with Roughness = 1.0, Metallic = 0.0, Specular = 0.0. The Base Color input uses the era palette color directly. No texture maps, no normal maps. A Geometry node (Normal > Dot Product > Facing) drives a MixRGB node that blends between the era's Shadow and Highlight colors.

Metal Material: Principled BSDF shader with Roughness = 0.4, Metallic = 1.0, Specular = 0.5. The Base Color uses the era's Stone/Earth color with a simple two-tone gradient simulating metallic reflection.

Used exclusively for weapons, armor plates, and metallic structural elements.

Emission Material: Emission shader used sparingly for glowing elements (boss auras, tech tower LEDs, UI indicators). Emission color uses the era's Accent color at strength 2.0. Overuse of emission materials breaks the cel-shaded aesthetic and is prohibited.

Transparency Material: Principled BSDF with Alpha = 0.6 for translucent elements (dragon wings, water surfaces, smoke effects). Blend mode set to Alpha Blend. No refraction, no subsurface scattering.

11.3 Outline Rendering (Freestyle)

All gameplay assets use Blender's Freestyle line rendering system to produce the distinctive dark outlines required by the art direction. Freestyle is configured as a render layer post-processing effect.

- Edge type: Silhouette + Border + Crease (all three enabled)
- Crease angle threshold: 60 degrees
- Line thickness: 2.0 pixels for standard assets, 3.0 pixels for character models
- Line color: #1A1A1A (near-black)
- No line thickness modulation along stroke; uniform width throughout

Exclusion Rules: Ground planes, shadow catchers, and background elements do not receive Freestyle outlines. Only gameplay objects are outlined. In scenes with multiple objects, use Freestyle Collection masks to restrict outline rendering.

11.4 Polycount Budgets

Polygon budgets are set for isometric sprite rendering, not real-time 3D display. Models can afford higher poly counts because they are rendered once and displayed as static images. However, reasonable budgets keep file sizes manageable and render times short.

Asset Type	Target Polycount	Max Polycount	LOD Levels
Tower (standard)	500 - 1,500 tris	2,500 tris	1 (LOD0 only)
Tower (hero/upgrade 3)	1,500 - 3,000 tris	5,000 tris	1 (LOD0 only)
Enemy (infantry)	800 - 1,500 tris	2,500 tris	1 (LOD0 only)
Enemy (cavalry)	1,500 - 3,000 tris	5,000 tris	1 (LOD0 only)
Enemy (siege)	2,000 - 4,000 tris	6,000 tris	1 (LOD0 only)
Enemy (boss)	3,000 - 6,000 tris	10,000 tris	1 (LOD0 only)
Hero (portrait)	3,000 - 5,000 tris	8,000 tris	N/A (portrait only)
Hero (in-game)	2,000 - 4,000 tris	6,000 tris	1 (LOD0 only)
Map (terrain tile)	200 - 500 tris	1,000 tris	1 (LOD0 only)

Prop (environmental)	100 - 300 tris	500 tris	1 (LOD0 only)

Table 11.1: Polycount Budgets

11.5 Sprite Sheet Export

Gameplay assets are exported as sprite sheets, not individual frames. Each sprite sheet contains all animation frames and rotation angles for a single asset arranged in a standardized grid layout.

11.5.1 Animation Frames

Asset Type	Animations	Frames	Notes
Tower (idle)	idle	1	Static pose; no animation needed
Tower (attack)	attack	4-6	Firing/recoil cycle; loop at last frame
Enemy (walk)	walk	6-8	Standard walk cycle; loop seamlessly
Enemy (attack)	attack	4-6	Attack animation; return to walk
Enemy (death)	death	4-6	Death/fade; do not loop
Hero (idle)	idle	4-6	Breathing/presence animation
Hero (ability)	ability	6-10	Ability activation; return to idle
Boss (phase 1)	attack_1	6-8	First phase attack pattern
Boss (phase 2)	attack_2	6-8	Second phase (triggered at HP threshold)

Table 11.2: Animation Frame Specifications

11.5.2 Render Resolution

All gameplay sprites are rendered at 256x256 pixels per frame and downscaled to the target display resolution in Godot. This 4x oversampling provides clean edges when displayed at 64x64 or 128x128. Portrait renders use 512x512 source resolution. Map background tiles use 1024x1024 source resolution, split into 256x256 tiles for Godot import.

Render settings: Output format = PNG with RGBA (alpha channel). Color depth = 8-bit per channel (sRGB). Compression = PNG lossless. No post-processing effects. Anti-aliasing = 8x multisample. The render layer outputs are composited using Blender's compositor only to combine the Freestyle outline layer with the color render layer.

11.5.3 Sprite Sheet Layout

Sprite sheets follow a standardized grid layout: frames are arranged horizontally (columns = time, rows = animation type). Each frame occupies a square cell whose size equals the render resolution (256x256).

The sheet header (first row, first column) is left blank as a padding pixel. Sheet naming follows the file naming convention in Section 13. For example: tower_archer_era00_spritesheet.png.

11.6 Blender Project Organization

The Blender project follows a strict file organization structure to maintain consistency across the solo developer workflow.

```
/assets/blender/library/ - Shared materials, lights, cameras, armatures
/assets/blender/era_00_colchis/ - Era 0 models, scenes, renders
/assets/blender/era_01_iberia/ - Era 1 models, scenes, renders
/assets/blender/era_02_invasions/ - Era 2 models, scenes, renders
/assets/blender/era_03_golden_age/ - Era 3 models, scenes, renders
/assets/blender/era_04_mongol/ - Era 4 models, scenes, renders
/assets/blender/era_05_empires/ - Era 5 models, scenes, renders
/assets/blender/era_06_russian/ - Era 6 models, scenes, renders
/assets/blender/era_07_republic/ - Era 7 models, scenes, renders
/assets/blender/era_08_soviet/ - Era 8 models, scenes, renders
/assets/blender/era_09_modern/ - Era 9 models, scenes, renders
/assets/blender/reference/ - AI concept images, historical photos
/assets/sprites/ - Final rendered sprite sheets
```

12. Scroll and Mastery Visual Design

Mastery scrolls are ERC-721 NFTs that provide permanent passive bonuses to the player's account. Their visual design must convey value, rarity, and cultural significance while remaining readable at the small sizes used in the marketplace and inventory UI. Scrolls are rendered in Blender using the same orthographic camera and cel-shading setup as gameplay assets.

12.1 Scroll Appearance

Each mastery scroll is depicted as a rolled parchment or Georgian-style manuscript page. The scroll body uses the era-appropriate Stone/Earth color for the parchment material, with the era's Accent color used for wax seals, decorative borders, and the scroll type icon. The scroll text area contains the scroll's name and bonus description in a stylized Georgian-inspired font, rendered as a texture on the scroll model.

Scroll rarity is indicated through a subtle glow effect using the Emission material:

- Common scrolls have no glow.
- Rare scrolls use the era's Highlight color at 0.5 emission strength.
- Epic scrolls use the era's Accent color at 1.0 emission strength.

- Legendary scrolls use the Global UI Accent (#E67E22) at 2.0 emission strength, making them visually unmistakable in any collection view.

12.2 Mastery Badge Display

When a player has 3 stacked copies of the same scroll (the maximum), a mastery badge appears on the scroll icon in the inventory. The badge is a small circular emblem using the Global UI Success color (#27AE60) with a checkmark icon, rendered at 24x24 pixels. The badge is composited in the Godot engine as a UI overlay, not rendered in Blender.

13. File Naming Convention

All project files follow a strict naming convention that encodes the asset type, era, variant, and version. This convention applies to Blender source files (.blend), rendered sprite outputs (.png), and Godot import files (.tres, .tscn). Consistent naming enables automated asset processing scripts and prevents naming conflicts across 200 levels and 10 eras.

13.1 Blender Source Files

Format: [type]_[name]_era[XX]_[variant].blend

- tower_archer_era00_base.blend
- enemy_infantry_era05_ottoman.blend
- hero_david_era03_portrait.blend
- map_level03_era02_siege.blend

Type Prefixes: tower_, enemy_, hero_, prop_, map_

13.2 Rendered Sprite Files

Format: [type]_[name]_era[XX]_[variant]_spritesheet.png

Portrait Format: hero_[name]_era[XX]_portrait.png

Map Format: map_level[XXX]_era[XX]_bg.png

- tower_archer_era00_base_spritesheet.png
- hero_tamar_era03_portrait.png
- map_level045_era02_bg.png

13.3 Material and Palette Files

Era Palette: palette_era[XX]_[name].hex (loaded into Blender's color palette system)

Material Library: materials_era[XX]_[name].blend (linked into era project files)

Example: palette_era03_golden_age.hex, materials_era03_golden_age.blend

13.4 Naming Rules

- All lowercase, no spaces, no special characters except underscores
- Era numbers are zero-padded two digits: era00 through era09
- Level numbers are zero-padded three digits: level001 through level200
- Variant names are single words: base, alternate, damaged, winter, etc.
- Version suffix for iterative updates: _v2, _v3 (appended before file extension)

14. Consistency Checklist

Every asset must pass this checklist before being approved for production rendering. The checklist is divided into visual consistency (how the asset looks) and pipeline consistency (how the asset is built). Each check corresponds to a specific rule in this style guide.

14.1 Visual Consistency

Check	Reference	Priority	Method
Era palette colors match exactly	Section 3	Critical	Eyedropper against .hex palette
Shadow direction lower-right 45 degrees	Section 1.3	Critical	Visual inspection of render
Outline weight 2px (3px characters)	Section 1.3	Critical	Pixel measurement at 100% zoom
Color fills flat (no gradients except metal)	Section 1.3	Critical	Visual inspection of material
Silhouette matches tower type reference	Section 4	Critical	Black silhouette overlay comparison
Proportions match 10-unit system	Section 5	Critical	Blender unit measurement
Era skin does not alter base silhouette	Section 4.2	Critical	Overlay comparison
Visible at 64x64 target resolution	Section 4.2	High	Downscaled render in Godot
Enemy visually distinct from towers	Section 7	Critical	Side-by-side at gameplay scale
Cultural sensitivity review passed	Section 7.3	Critical	Manual review

Table 14.1: Visual Consistency Checklist

14.2 Pipeline Consistency

Check	Reference	Priority	Method
File naming follows convention	Section 13	Critical	Automated naming script
Materials use shared era palette .hex	Section 11.2	Critical	Blender material verification
Freestyle line set matches standard	Section 11.3	Critical	Freestyle settings comparison
Render resolution matches spec	Section 11.5.2	Critical	Output dimensions verification
Alpha channel present in PNG	Section 11.5.2	Critical	PNG metadata check
Polycount within budget	Section 11.4	High	Blender triangle count
Sprite sheet layout follows grid	Section 11.5.3	High	Frame grid alignment
Scene uses single Sun light only	Section 11.1	Critical	Outliner light list check
Orthographic camera correct rotation	Section 11.1	Critical	Camera transform values
Concept brief document exists	Section 10.3	Medium	File existence check

Table 14.2: Pipeline Consistency Checklist

15. Hermes Collaboration Scope

Hermes (the AI art collaborator) assists with the Concept Reference stage of the pipeline (Section 10) and initial Blender blocking. Hermes does not produce final game-ready assets; all final assets are produced by the developer through the Blender pipeline defined in Section 11.

15.1 Hermes Responsibilities

- Generate AI concept images using the prompt architecture (Section 10.1) for mood board exploration
- Create concept brief documents that translate AI concepts into Blender modeling specifications
- Suggest material and color assignments based on era palette research
- Provide initial Blender blocking geometry (basic shapes scaled to 10-unit proportion system)
- Review rendered sprite output against the consistency checklist (Section 14) and flag issues
- Suggest era-appropriate architectural and military details based on historical research

15.2 Hermes Limitations

- Hermes cannot directly edit .blend files or run Blender renders

- Hermes cannot produce final sprite sheets; the developer renders all production assets
- Hermes cannot modify the Freestyle line set or material system configuration
- Hermes does not have authority to change era palette colors or silhouette definitions
- All Hermes suggestions are advisory; the developer makes all final art direction decisions

15.3 Priority Work Order

The following work items are prioritized for Hermes collaboration, in order of dependency:

- Priority 1: Tower Silhouette Reference Sheet - Render all 10 tower types as flat black silhouettes on white background at 256x256px
- Priority 2: Era 0 Palette Validation - Generate concept images using the Era 0 palette to validate the color scheme
- Priority 3: Character Proportion Reference Armature - Create a Blender armature with the 10-unit proportion system and all character type variants
- Priority 4: Shared Scene Template - Create a Blender scene file with the standardized camera, lighting, and Freestyle setup
- Priority 5: Era Concept Batches - Generate concept reference sets for each era (10-15 images per era)

16. Version History

v2.1 - Blender Pipeline Update

This version incorporates the transition from AI-only asset production to a Blender-based 3D-to-2D sprite pipeline. The visual style and art direction remain unchanged; only the production methodology and pipeline specifications are new. Key changes include:

- Section 1.2 (NEW): Production Pipeline - Blender to 2D Sprites workflow documentation
- Section 1.3 (UPDATED): Rendering Rules - Added Blender-specific implementation details
- Section 5 (UPDATED): Character Proportions - Added Blender armature and unit system references
- Section 6.2 (NEW): Portrait Render Setup - Standardized Blender scene for hero portraits
- Section 10 (REPLACED): AI Concept Reference System - AI generation downgraded to concept reference stage
- Section 11 (NEW): Blender Pipeline - Complete production pipeline specifications
- Section 12.1 (UPDATED): Scroll Appearance - Added Blender material and emission specifications

- Section 13 (UPDATED): File Naming - Added .blend source file naming and material/palette naming
- Section 14 (UPDATED): Consistency Checklist - Split into Visual and Pipeline subsections
- Section 15 (UPDATED): Hermes Scope - Updated deliverables for Blender pipeline

v2.0 - Initial Release

Initial Art Style Guide release aligned with Game Design Bible v3.0. Established the 10-era color palette system, tower silhouette guidelines, character proportion system, hero portrait guidelines, enemy design standards, map design rules, and UI design language. Defined the AI prompt architecture as the primary production tool. Included file naming convention, consistency checklist, and Hermes collaboration scope.